# COM 5506

*L 2 Node Centrality*

# Social Network Analysis for Communication

SEM B 2025/2026

Fei Shen, City University of Hong Kong

# Recapitulation

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- Concepts:
  - nodes, edges
  - degree, edge weight, path, distance, diameter, average path length
  - components, ego centric network , density
  - adjacency matrix, edge list
- Understand:
  - Matrix
  - Attribute data and relational data structure
  - Matrix permutation
  - Matrix multiplication

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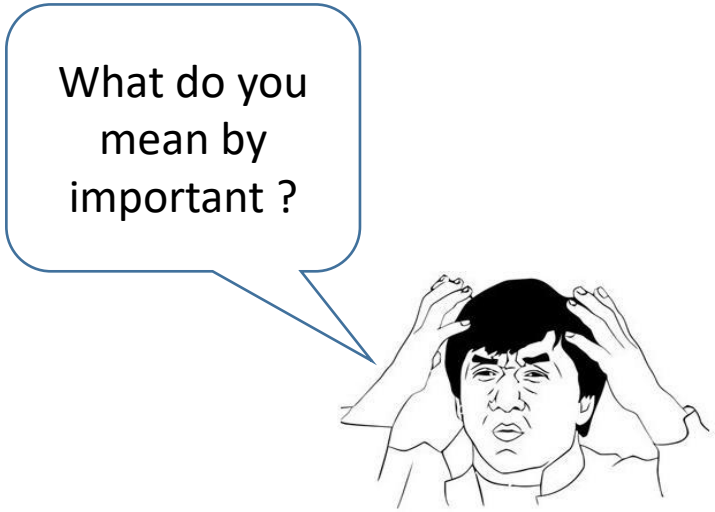
Who are the most important/influential person?

What is the key node in the infrastructure?



# I know a lot of people!

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- *Degree centrality (DC)*
- A measure of "popularity" of node i

$$DC_i = \frac{D_i}{n-1}$$

where $D_i$ is the degree of node i

- DC indicates how well the node is connected within the graph

# Degree Centrality (DC)

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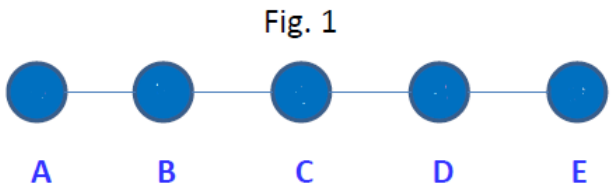


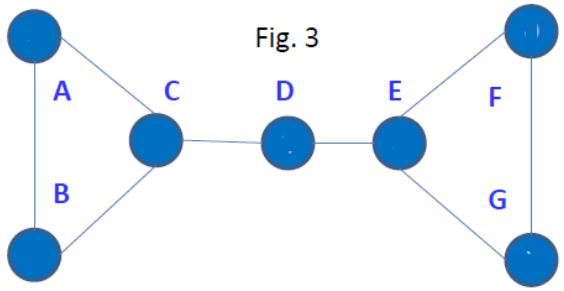


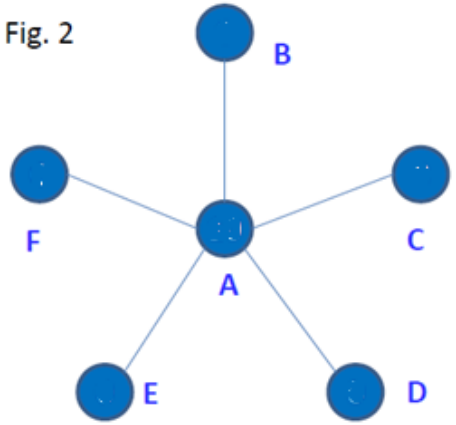


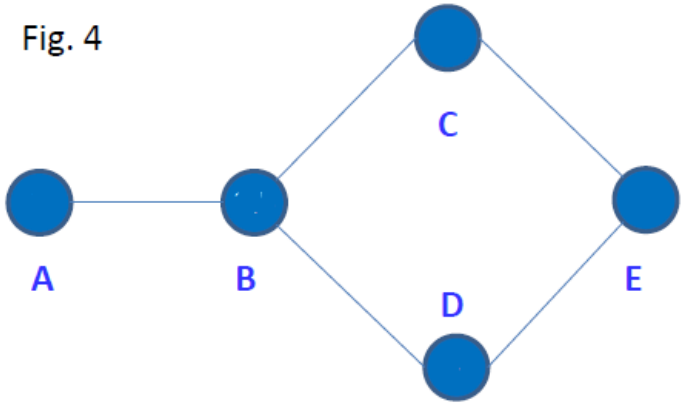


| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| Fig1 | 1 | 2 | 2 | 2 | 1 | -- | -- |
| Fig2 | 5 | 1 | 1 | 1 | 1 | 1 | -- |
| Fig3 | 2 | 2 | 3 | 2 | 3 | 2 | 2 |
| Fig4 | 1 | 3 | 2 | 2 | 2 | -- | -- |

# Problem with degree centrality

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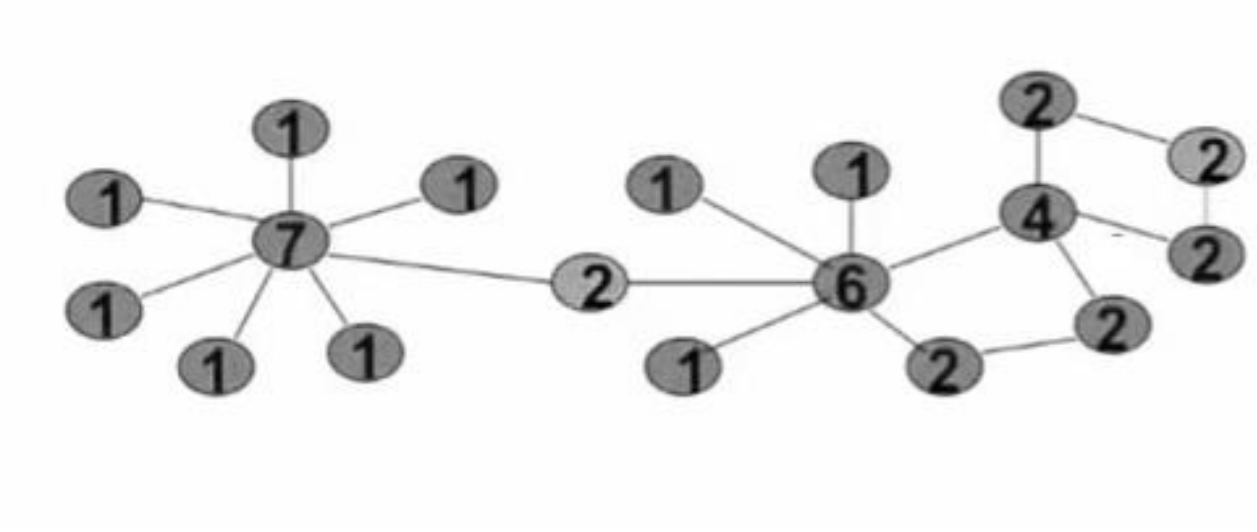
- Failure of degree centrality to capture reach of a node
- Failure of degree centrality to consider the existence of subcomponents



Harvard Business Review

NETWORKING

# Your Network's Structure Matters More than Its Size

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by Kelsey Libert

FEBRUARY 23, 2016

- "A recent study by Schlesinger Associates for Augure found that 75% of marketers consider finding the right influencers the most challenging aspect of an influencer marketing strategy. Perhaps that's due to a misguided approach in which the size of someone's following is treated as the primary benchmark of their influence."
- New research shows that you only need handful of influencers to give the impression that everyone is talking about your brand.

# Being an influential node

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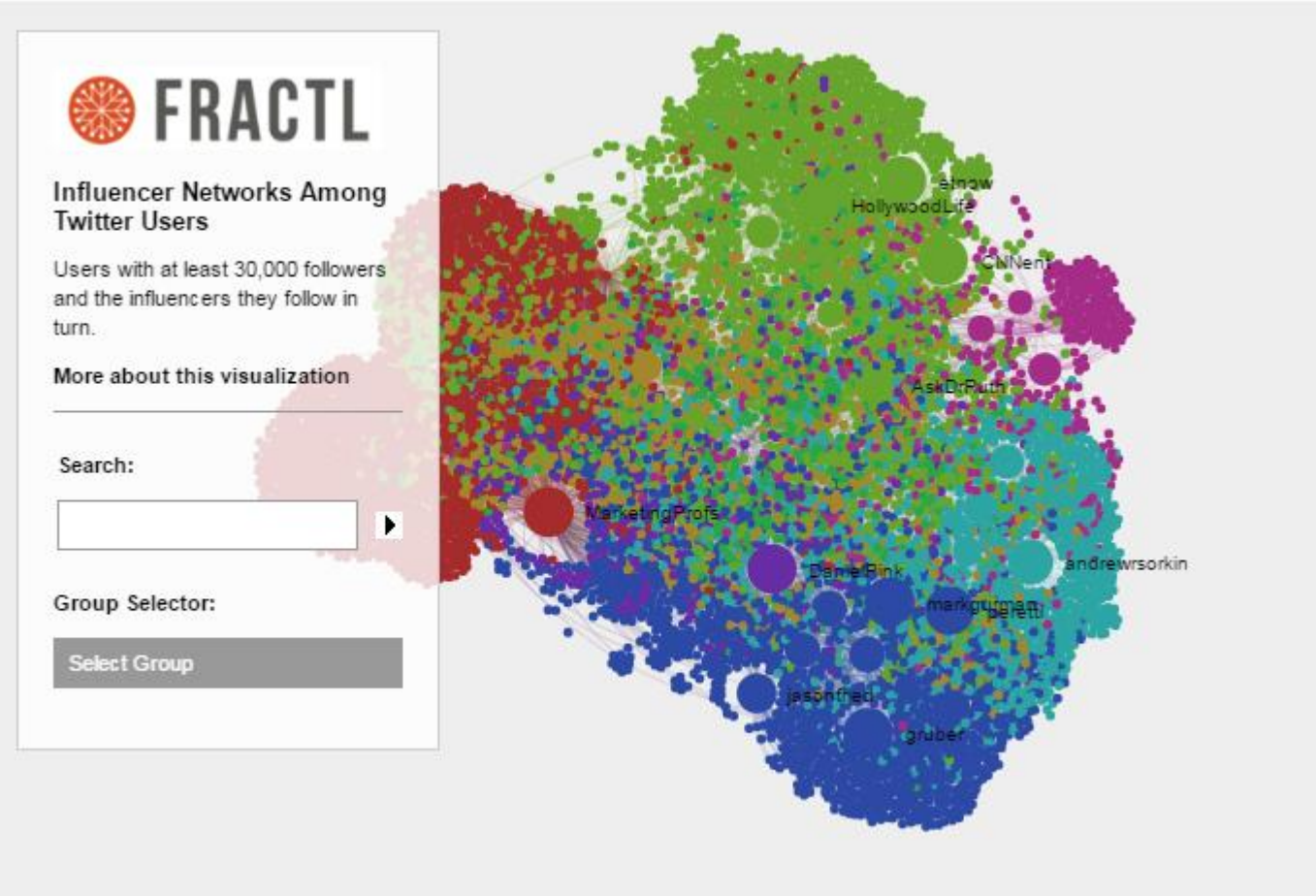
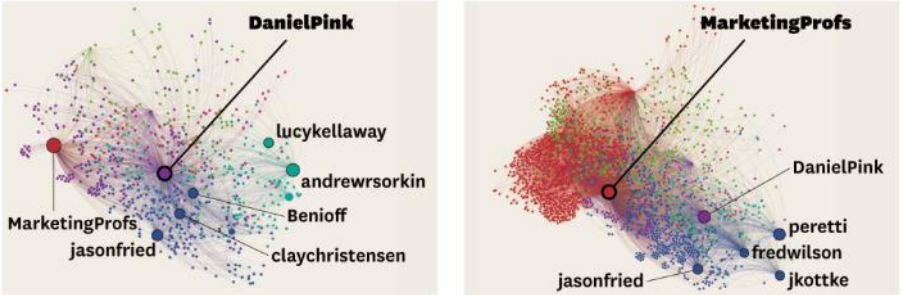
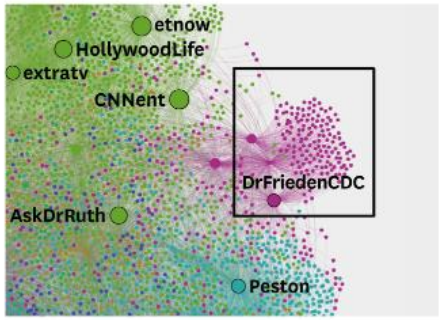



**What Betweenness Centrality Looks Like**




SOURCE FRACTL, BASED ON AN ANALYSIS OF TWITTER INFLUENCERS © HBR.ORG

**What Network Closeness Looks Like**




SOURCE FRACTL, BASED ON AN ANALYSIS OF TWITTER INFLUENCERS © HBR.ORG

# I know different groups of people!

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- *Betweenness Centrality (BC)*
- A measure of the "intermediacy" role of node

$$C_B(i) = \sum_{j<k} g_{jk}(i) / g_{jk}$$

$$C_B'(i) = C_B(i) / [(n-1)(n-2)/2]$$

number of pairs of vertices excluding the vertex itself

where $g_{jk}$ is the number of shortest paths between all pairs of j and k (j≠i≠k) and $g_{jk}(i)$ is the number of the shortest paths that pass through i.

- BC indicates how critical the node is to the flow of information or resource in the group.

# Betweenness Centrality (BC)

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Fig. 1

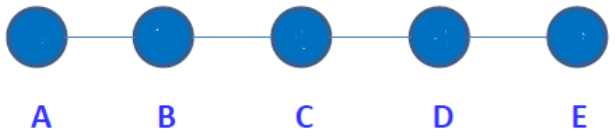



Fig. 3

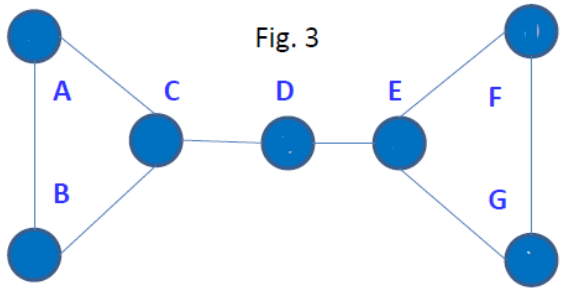



Fig. 2

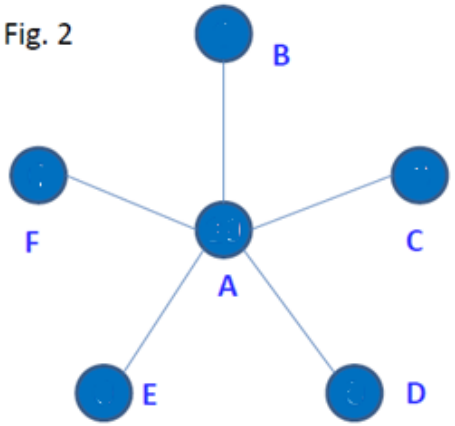
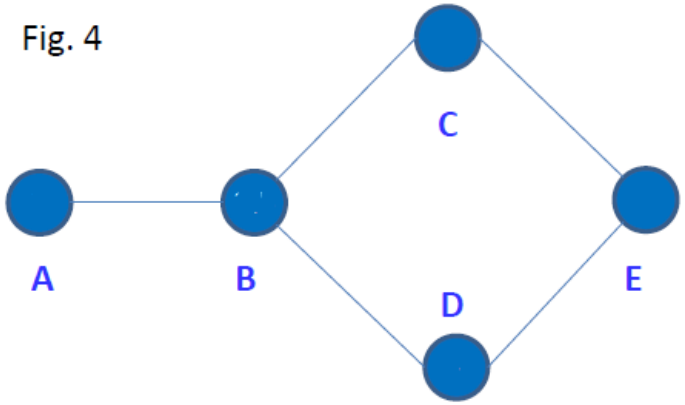


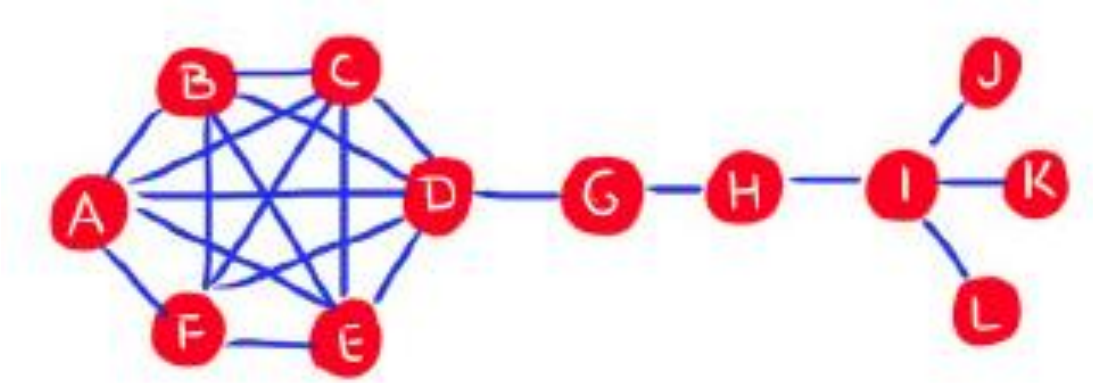

Fig. 4




| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| Fig1 | 0 | 3 | 4 | 3 | 0 | -- | -- |
| Fig2 | 10 | 0 | 0 | 0 | 0 | 0 | -- |
| Fig3 | 0 | 0 | 8 | 9 | 8 | 0 | 0 |
| Fig4 | 0 | 3.5 | 1 | 1 | 0.5 | -- | -- |

# Q: betweenness and degree

- Find a node that has high betweenness but low degree
- Find a node that has low betweenness but high degree



# I get to know all people easily!

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- *Closeness Centrality* (CC)
- A measure of "proximity" of node i to other nodes:

Correction: This should be multiplication *

$$C_c(i) = \left[\sum_{j=1}^{N} d(i,j)\right]^{-1} \qquad C_C'(i) = (C_C(i))/(N-1)$$

where d(i,j) is the shortest path length between i and j. It is the total shortest path lengths of i to all other nodes.

- CC indicates how quickly information travels between node i and anyone else in the graph.

# Example

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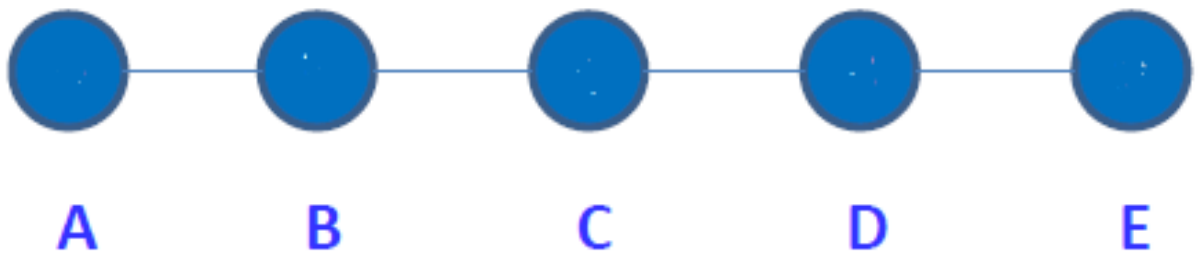
Fig. 1




$$C_c'(A) = \left[ \frac{\sum_{j=1}^{N} d(A,j)}{N-1} \right]^{-1} = \left[ \frac{1+2+3+4}{4} \right]^{-1} = \left[ \frac{10}{4} \right]^{-1} = 0.4$$

# Closeness Centrality (CC)

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Fig. 1

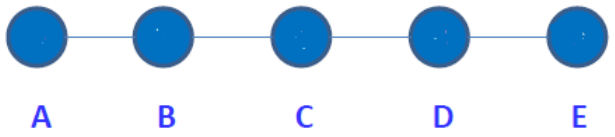



Fig. 3

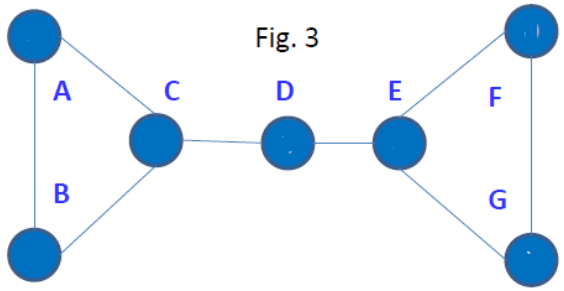



Fig. 2

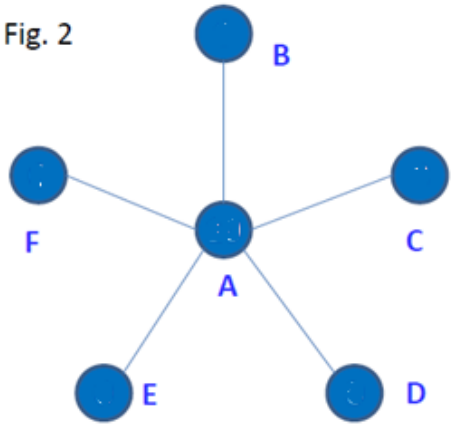
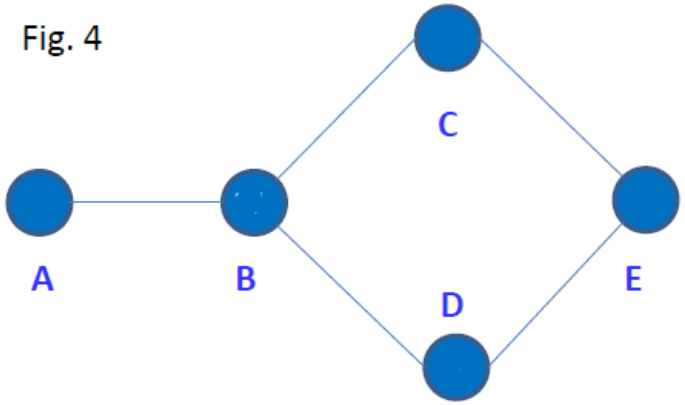



Fig. 4




| | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| Fig1 | 0.4 | .57 | .67 | .57 | 0.4 | -- | -- |
| Fig2 | 1 | .56 | .56 | .56 | .56 | .56 | -- |
| Fig3 | .40 | .40 | .55 | .60 | .55 | .40 | .40 |
| Fig4 | .50 | .80 | .67 | .67 | .57 | -- | -- |

# I know people from the royal family!

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- A measure of "popularity of the neighbors" of node i:

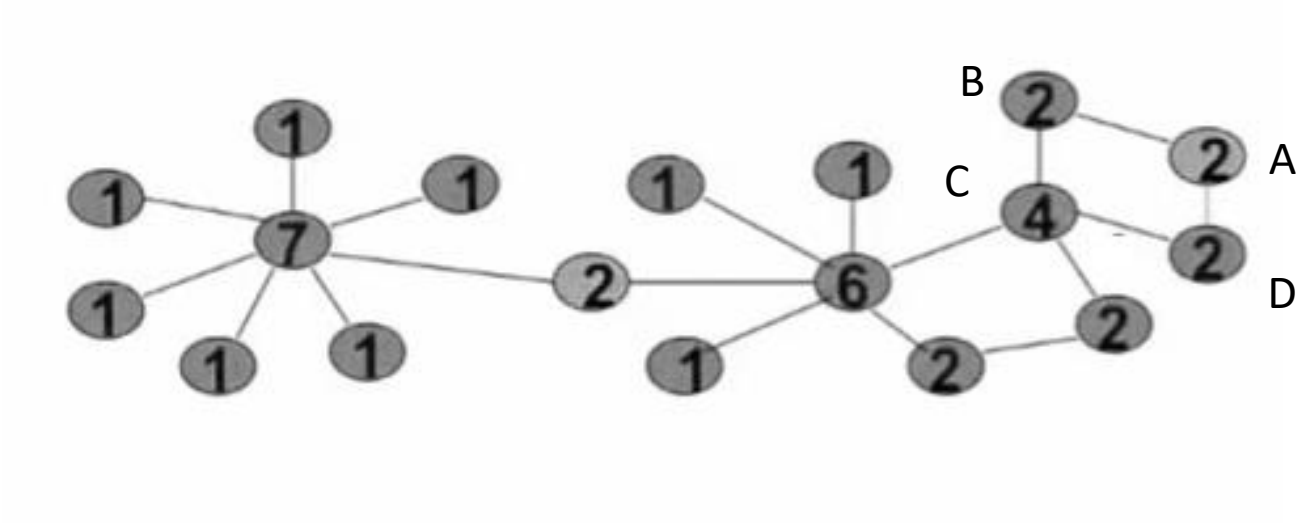
$$EC_i = \sum_{j=1}^{n-1} EC_j$$

where j is a neighbor of i, and $EC_j$ is the eigenvector of j, which depends on the eigenvector of its neighbors, which depends in turn on the EC of its neighbors

- EC measures how well connected node i is to the parts of the network with the greatest connectivity.

# Eigenvalue Centrality (EC)

- Finding *eigenvalue centrality* conceptually is an iterative process, making it impossible to count and to eyeball the number.
- For instance, node A's importance depends on B and D, and so on, but node B's importance, depends on C and A, and so on.
- It then becomes a chicken and egg problem, but luckily linear equation provides us solutions and this is exactly why it is called eigenvalue centrality.



# Eigenvalue Centrality (EC)

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- Definition

$$EC_i = \sum_{j=1}^{n-1} EC_j$$

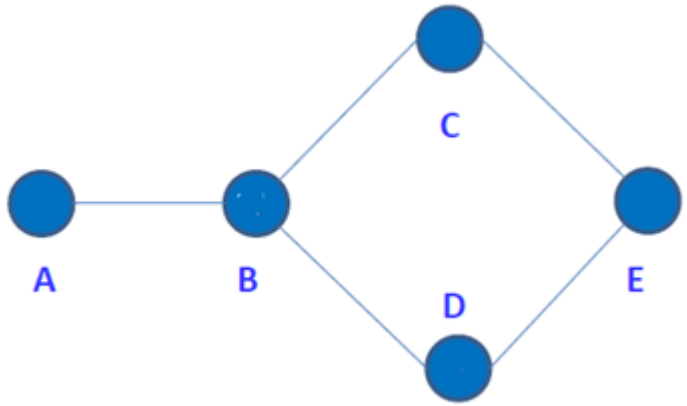
where i and j are any node in the graph

Why this is called Eigenvalue centrality and what is Eigen value?

- If λc = G*c, where λ, is called eigenvalue, and c is called eigenvector.

# Equation roughly illustrated

$$\begin{bmatrix} va \\ vb \\ vc \\ vd \\ ve \end{bmatrix} = \frac{1}{\lambda} \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} va \\ vb \\ vc \\ vd \\ ve \end{bmatrix}$$



# Eigenvalue and Eigenvector

- Matrix transformation

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} =$$

# The Google centrality measure: Page Rank for directed network

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- To some extent, finding the most “important” people is similar to finding the most “important” webpage.
- Google used the idea of eigenvalue centrality and invented the “page rank” algorithm.
- Initially developed by Larry Page and Serge Brink to rank webpages in search engines

# Page Rank algorithm

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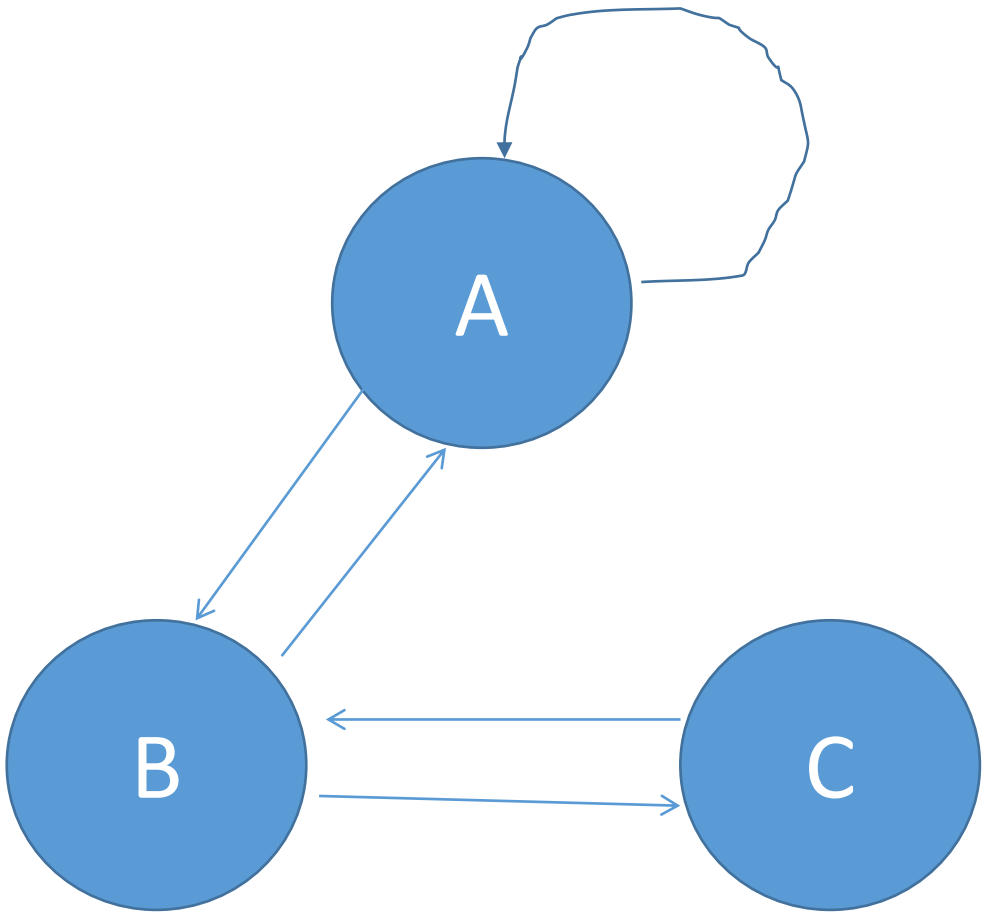
- General rules:
  - The more pages linking to you, the more important you are (degree centrality)
  - Links from pages with more in-degree linkage will boost your rank more than links from isolated pages (eigenvalue centrality)
  - The importance passed on through a link will be diluted if there are many other out-degree links coming out from that page (this is new. different than eigenvalue centrality)
  - Links exist a long time matter more

# PageRank Illustrated




| | A | B | C |
|---|---|---|---|
| A | ½ | ½ | 0 |
| B | ½ | 0 | 1 |
| C | 0 | ½ | 0 |

# PageRank illustrated

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- Power iteration method
  - Set $P_i = 1/n$
  - $P_i = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \end{bmatrix}$ $G = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 0 & 1 \\ 0 & 1/2 & 0 \end{bmatrix}$
  - $P_i \times G$ until $P$ converges

| | | | | | |
|---|---|---|---|---|---|
| 1/3 | 1/3 | 5/12 | 9/24 | | 6/15 |
| 1/3 | 3/6 | 1/3 | 11/24 | … | 6/15 |
| 1/3 | 1/6 | 3/12 | 1/6 | | 3/15 |

Iteration 0, 1, 2, …

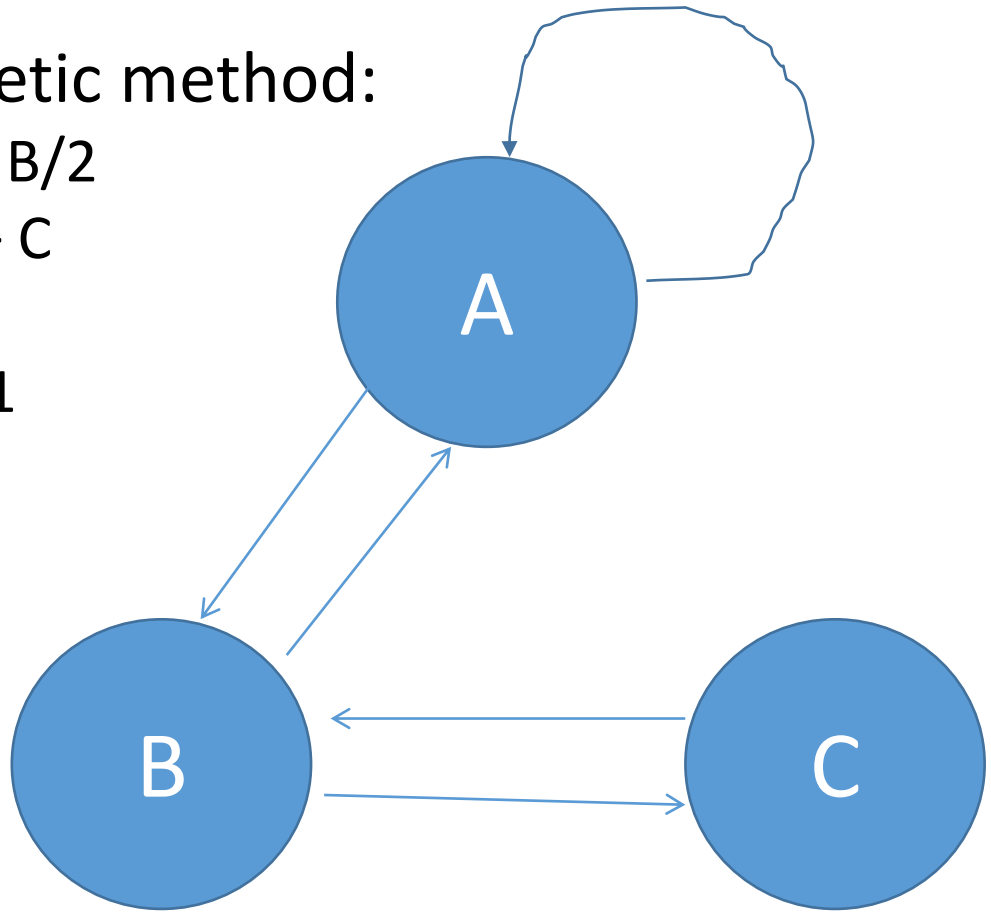
Arithmetic method:

A= A/2+ B/2

B = A/2+ C

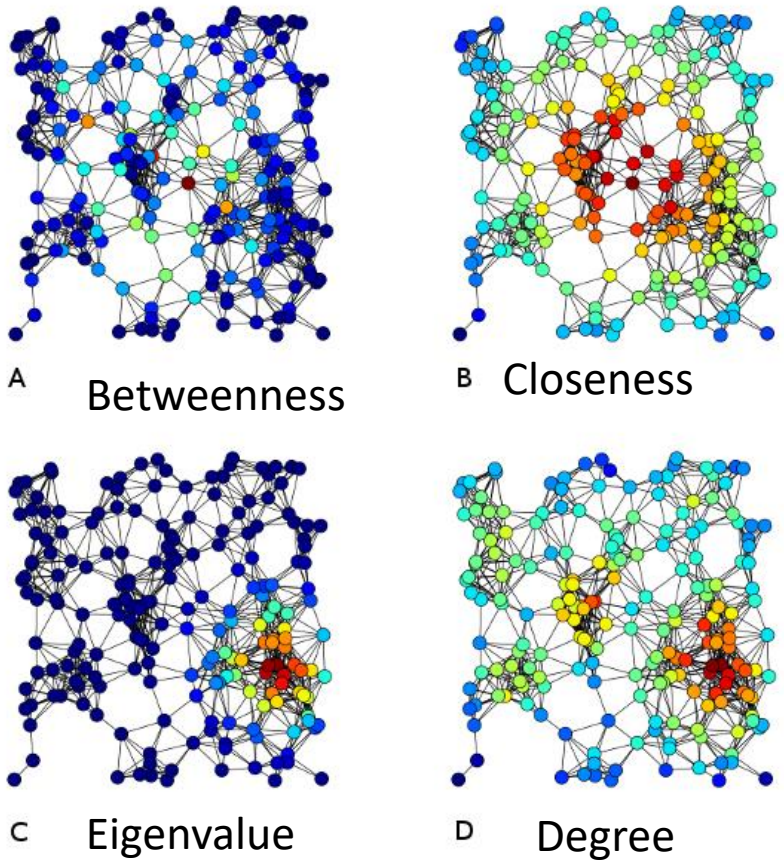
C= B/2

A+B+C=1



# Centrality measures in summary

- Degree – connectedness
- Closeness – ease of reaching other nodes
- Betweenness – importance as an intermediary, connector
- Eigenvector – not what you know, but who you know
- The four measures are not necessarily correlated!



# Centralization: network level feature

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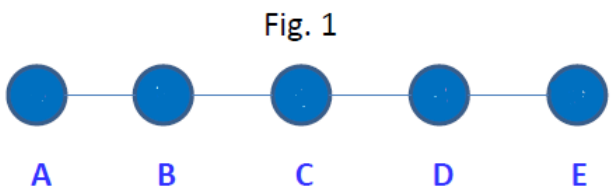
- How much variation is there in the centrality scores among the nodes?
- Freeman's general formula for centralization

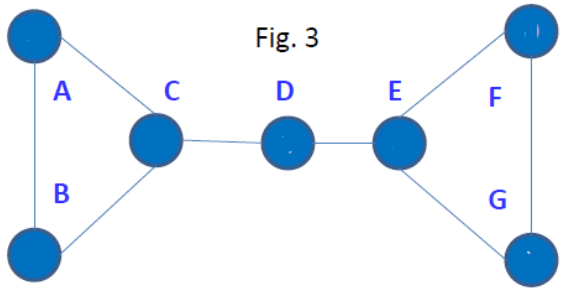
maximum value in the network

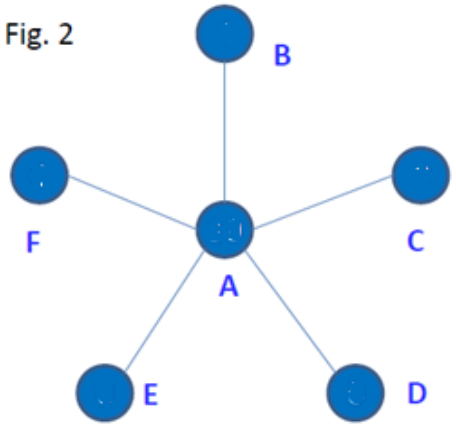
$$C_D = \frac{\sum_{i=1}^{g} \left[ C_D(n^*) - C_D(i) \right]}{[(N-1)(N-2)]}$$

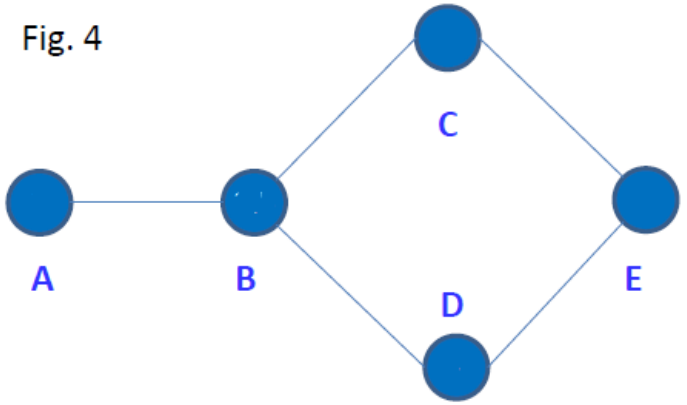
- Of course, other metrics are good to use, e.g. gini coefficient or standard deviation

# Degree centralization examples




- C1=0.167
- C2=1
- C3= 0.167

# Lab session 2

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- “Python is a widely used high-level, general-purpose, interpreted, dynamic programming language.”
  - Readability
  - Open source
- “I chose Python as a working title for the project, being in a slightly irreverent mood (and a big fan of Monty Python's Flying Circus).” - Guido van Rossum

# Python distributors

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- WinPython, Python(XY), Anaconda, etc.
- For this course, we choose Anaconda, for a number of reasons
  - support both Windows and Mac
  - include Ipython Notebook, Spyder, and most pre-installed libraries
- Install the most recent version.

ANACONDA
Powered by Continuum Analytics

# IP[y]: Notebook

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- The IPython Notebook is now known as the Jupyter Notebook. It is an interactive computational environment, in which you can combine code execution, rich text, mathematics, plots and rich media.
- The default working directory is your user/’name’ directory under windows…